rag -- 2026-11-11

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P5] PatternBloom: Empowering Agentic RAG with Externalized RL-Distilled Graph Patterns -- PatternBloom demonstrates a concrete, empirically validated advance in grounding and reasoning depth for RAG, achieving state-of-the-art multi-hop QA with a modest 7B model and detailed ablations that clarify the impact of its graph-pattern memory and reward design.

Themes

These papers converge on improving the retrieval component of RAG systems, but they attack different bottlenecks. Several works (P2, P5) emphasize grounding and faithfulness: P2 introduces two-sided conformal evidence sets that give finite-sample coverage guarantees for both supporting and refuting passages, showing near-nominal coverage on the VitaminC benchmark and a decision accuracy of 0.802 on FEVER [P2]. P5 externalizes procedural reasoning into a frozen Graph Pattern Memory and uses a pattern-augmented reward to boost multi-hop QA, reporting a 15.0-point OOD exact-match gain over strong baselines with a 7B model [P5]. By contrast, P4 tackles the token-budget limitation of RAG by compressing retrieved documents into a 16-token visual prefix, achieving the highest compression-aware performance on LongBench-v2 (+5.17 points) and strong CAP scores on BAMBOO and LooGLE-v2 [P4]. P1 proposes an activation-guided graph retrieval control layer that dynamically expands a memory graph based on activation scores, but provides only analytical cost bounds and no empirical validation [P1]. P3 shifts the focus to non-textual modalities, presenting Retrieval-Augmented Guidance (RAGu) that warps out-of-domain demonstrations for a frozen vision-language-action model, raising zero-shot success from 20 % to 80 % without parameter updates [P3]. Together, the set highlights three recurring design axes: (1) richer evidence calibration (P2), (2) structured procedural memory (P5) and graph-based retrieval (P1), (3) token-efficient multimodal retrieval (P4) and retrieval-driven test-time adaptation for embodied agents (P3). Evaluation gaps remain--most papers test on narrow benchmarks (e.g., VitaminC, FEVER, LongBench, specific robot tasks) and do not provide cross-domain or open-ended generation results, leaving a need for unified, multimodal RAG benchmarks.

5 theme sentence(s) checked against the paper summaries.

At a glance

[P1] Activation-Guided Graph Retrieval for Cognitive Memory Reconstruction in Language Model Systems · relevance 0.40
The paper proposes an activation-guided graph retrieval (AGGR) architecture to make memory availability stateful in long-lived language-model systems, but provides only analytical specifications and no empirical results.

[P2] Two-Sided Conformal Evidence Sets for Retrieval-Augmented Generation · relevance 0.38
The paper introduces Conformal Evidence Sets (CES), a two-sided conformal calibration method for retrieval-augmented generation that provides finite-sample coverage guarantees for both supporting and refuting evidence, and demonstrates near-nominal coverage on benchmark datasets while highlighting distinct trade-offs between evidence coverage and verdict accuracy.

[P3] Zero-Shot VLA Adaptation with Retrieved Guidance · relevance 0.38
RAGu retrieves and warps out-of-domain demonstrations to guide a frozen Vision-Language-Action model at test time, boosting zero-shot success from 20 % to 80 % without any parameter updates.

[P4] OpticalRAG: Pixel-Space Compression for Token-Efficient Retrieval-Augmented Generation · relevance 0.36

OpticalRAG renders documents as images, compresses them into visual tokens, and injects only 16 tokens into a LLM, achieving the best compression-aware accuracy on several long-context QA benchmarks while improving the backbone score by 5.17 points on LongBench-v2 [1][3].

[P5] PatternBloom: Empowering Agentic RAG with Externalized RL-Distilled Graph Patterns · relevance 0.35
PatternBloom introduces a two-stage RL framework that externalizes procedural reasoning as graph pattern memory, achieving state-of-the-art multi-hop QA performance with a 7B model.

[P1] Activation-Guided Graph Retrieval for Cognitive Memory Reconstruction in Language Model Systems

Marvin Danig · arXiv 2611.13127v1 · https://arxiv.org/abs/2611.13127v1 · relevance 0.40

TL;DR: The paper proposes an activation-guided graph retrieval (AGGR) architecture to make memory availability stateful in long-lived language-model systems, but provides only analytical specifications and no empirical results.

Problem
Current retrieval-augmented generation systems generally select context by query-conditioned relevance, and this relevance score does not fully specify whether a memory should influence inference now [1][2].

Method
The author defines AGGR as a control layer that (1) seeds an authorized memory graph with request and goal cues, (2) computes activation scores for candidate entities, (3) allocates a bounded traversal frontier using activation and fan-aware attenuation, (4) expands paths recursively as retrieved entities emit new cues, and (5) reconstructs a typed evidence graph before generation [2][3][4]. The activation model combines a base-level term (Equation 4) with query relevance and other factors (Equation 5), while fan attenuation reduces the influence of high-degree cues (Equation 6-7).

Key results
- No benchmark measurements or empirical results are reported; the paper only offers analytic cost bounds [5][6].
- Traversal worst-case complexity is O(b h) node visits, and with a bounded frontier of size L the scoring cost is O(h L b log L) [5].
- Context token cost is approximated as T_context ~ |V_e|(u + ?) + |E_e| l, where |V_e| and |E_e| are numbers of node summaries and edge labels, respectively [5].
- An evaluation protocol is outlined, comparing six systems (lexical retrieval, dense/hybrid top-k, graph retrieval without activation, activation reranking, full AGGR, and a write-time consolidated-notes baseline), but no empirical data are provided [6].

Limitations
The paper reports no measurements, so all claimed benefits remain hypotheses until the proposed evaluation is executed [6]. Equations 4 and 5 are engineering approximations with uncalibrated weights, risking worse performance than plain similarity if set poorly [6]. The approach assumes a largely correct memory graph; mis-labelled edges can produce confident but incorrect causal chains [6]. Potential failure modes include amplification of stale, biased, or improperly authorized memories, requiring careful provenance, access-control testing, and auditable retrieval traces [7][8].

Sources

[1] arXiv 2611.13127v1.pdf, arXiv abstract

[2] arXiv 2611.13127v1.pdf, page 1

[3] arXiv 2611.13127v1.pdf, page 2

[4] arXiv 2611.13127v1.pdf, page 3

[5] arXiv 2611.13127v1.pdf, page 5

[6] arXiv 2611.13127v1.pdf, page 6

[7] arXiv 2611.13127v1.pdf, page 7

[8] arXiv 2611.13127v1.pdf, page 8

[P2] Two-Sided Conformal Evidence Sets for Retrieval-Augmented Generation

Ben Jenkins, Mihaela Cardei · arXiv 2611.10335v1 · https://arxiv.org/abs/2611.10335v1 · relevance 0.38

TL;DR: The paper introduces Conformal Evidence Sets (CES), a two-sided conformal calibration method for retrieval-augmented generation that provides finite-sample coverage guarantees for both supporting and refuting evidence, and demonstrates near-nominal coverage on benchmark datasets while highlighting distinct trade-offs between evidence coverage and verdict accuracy.

Problem
Existing calibrated grounding approaches for retrieval-augmented generation handle only supporting evidence and therefore miss refuting passages, conflating evidence coverage with claim-verification performance [1][2].

Method
The authors propose Conformal Evidence Sets (CES), a claim-level two-sided conformal formulation that constructs separate support and refutation sets with finite-sample inclusion guarantees (>= 1 - ?) under exchangeability [3][4]. An algorithm computes nonconformity scores from an NLI scorer, derives side-specific thresholds via order statistics, and applies a decision rule that returns "support", "refute", or "abstain" based on set emptiness and a dominance margin ? (set to 0 in experiments) [3][4]. Theoretical guarantees are given in Proposition 1, and the method is evaluated using a DeBERTa-v3 NLI model fine-tuned on MNLI, ANLI, and FEVER [4].

Key results
- On the contrastive VitaminC benchmark, CES tracks nominal coverage across ? ? {0.05, 0.10, 0.15, 0.20, 0.30} with maximum deviations of 3.0 pp for support and 5.4 pp for refutation [1].
- CES achieves essentially the same empirical coverage-size frontier as a test-tuned fixed-threshold heuristic, using about 3.26 passages on average at ? = 0.10 (existential coverage 0.921) versus 3.07 passages for the fixed-threshold baseline [5][6].
- Under FEVER distribution shift, both sides under-cover by up to 12.7 pp (largest miscoverage observed at ? = 0.20) [1][6].
- In FEVER, CES attains support coverage 0.833, refutation coverage 0.815, decision accuracy 0.802, and abstention precision 0.733 at ? = 0.10, matching top-1 NLI accuracy (0.805) while providing calibrated evidence on both sides [6].
- The one-sided conformal baseline never emits refutations and its decision accuracy drops to 0.577, highlighting the advantage of two-sided calibration [6].
Stress tests show that coverage remains stable under scorer and distractor perturbations, and degrades under transfer in a manner consistent with exchangeability violation [2].

Limitations
Coverage guarantees assume that relevant evidence is present in the retrieved pool and do not ensure factual correctness [7]. VitaminC is deliberately contrastive and contains no gold-NEI claims, so it overestimates the frequency of having evidence on both sides; FEVER and a live pilot reveal this limitation [7]. The FEVER grouping is dataset-specific, the NLI scorer inherits its training biases, and retrieval, generation, and decomposition steps lie outside the formal guarantee [7]. CES does not explicitly model claims that require jointly sufficient multi-passage evidence [7].

Sources

[1] arXiv 2611.10335v1.pdf, arXiv abstract

[2] arXiv 2611.10335v1.pdf, page 1

[3] arXiv 2611.10335v1.pdf, page 3

[4] arXiv 2611.10335v1.pdf, page 4

[5] arXiv 2611.10335v1.pdf, page 5

[6] arXiv 2611.10335v1.pdf, page 6

[7] arXiv 2611.10335v1.pdf, page 7

[P3] Zero-Shot VLA Adaptation with Retrieved Guidance

Subin Kim, Anh-Quan Pham, Dinesh Jayaraman · arXiv 2611.10894v1 · https://arxiv.org/abs/2611.10894v1 · relevance 0.38

TL;DR: RAGu retrieves and warps out-of-domain demonstrations to guide a frozen Vision-Language-Action model at test time, boosting zero-shot success from 20 % to 80 % without any parameter updates.

Problem
Pre-trained VLA models achieve strong manipulation performance but degrade sharply under modest domain shifts such as changes in camera viewpoint, background, object identity, or robot morphology [1]. Adapting them is difficult because (i) cross-domain visual and geometric differences exist, (ii) the closest trajectory in the source data may be incompatible with the policy, and (iii) the VLA's limited context window cannot easily absorb many retrieved examples [3].

Method
The authors propose Retrieval-Augmented Guidance (RAGu), a training-free framework that (1) calibrates geometry by warping source demonstrations into the target scene using semantic bounding-box correspondences [3][4]; (2) employs a coarse-to-fine, policy-aware retrieval pipeline that first prunes candidates with offline dynamic-time-warping and then selects the trajectory most compatible with the base VLA [2][3]; and (3) constructs an energy-based guidance field to steer the VLA's flow-matching denoising inference, leaving the model weights and context untouched [1][2].

Key results
- Real-world VLA success rates rise from 20 % to 80 % when RAGu is applied [1].
- In a cross-embodiment transfer (7-DoF Franka demonstrations guiding a 29-DoF Fourier GR-1 humanoid), using 100 retrieved candidates increases average success by +13.6 % [6].
- Experiments span four transfer settings (real-to-real, real-to-sim, sim-to-real, and arm-to-humanoid) with 10 rollouts for real-world tests and 25 for simulation [5].
- Larger retrieval pools further improve robustness, though exact quantitative gains are not detailed [7].

Limitations
RAGu relies on the source database containing demonstrations that roughly match the target task category, is still sensitive to camera placement and occlusions that can break semantic correspondences, and can only reshape motion within the existing VLA prior without expanding its capabilities [7].

Sources

[1] arXiv 2611.10894v1.pdf, arXiv abstract

[2] arXiv 2611.10894v1.pdf, page 1

[3] arXiv 2611.10894v1.pdf, page 2

[4] arXiv 2611.10894v1.pdf, page 3

[5] arXiv 2611.10894v1.pdf, page 3

[6] arXiv 2611.10894v1.pdf, page 5

[7] arXiv 2611.10894v1.pdf, page 6

[P4] OpticalRAG: Pixel-Space Compression for Token-Efficient Retrieval-Augmented Generation

Senhao Liu, Yuheng Zhang, Chunyu Wei, Yueguo Chen et al. · arXiv 2611.13675v1 · https://arxiv.org/abs/2611.13675v1 · relevance 0.36

TL;DR: OpticalRAG renders documents as images, compresses them into visual tokens, and injects only 16 tokens into a LLM, achieving the best compression-aware accuracy on several long-context QA benchmarks while improving the backbone score by 5.17 points on LongBench-v2 [1][3].

Problem
Retrieval-augmented generation is constrained by the limited token budget of large language models, making it expensive to feed long documents and causing aggressive textual compression to discard fine-grained information irreversibly [1].

Method
OpticalRAG renders each document page as an image and encodes it with a frozen vision encoder; a text query is projected into the same visual feature space (Encoder-Bridged Retrieval) without extra contrastive training, and a lightweight cross-attention compressor (Query-Driven Distillation) condenses the retrieved visual tokens into a query-conditioned prefix of as few as 16 tokens that are injected into Qwen2.5-7B-Instruct [1][2].

Key results
On LongBench-v2 OpticalRAG attains a raw score of 35.79, improving the closed-book backbone by 5.17 points while using only 16 injected tokens and achieving the highest CAP [3].
On BAMBOO, OpticalRAG reaches a raw score of 61.00 and the best CAP of 84.67, while RECOMP and LLMLingua-v2 achieve higher raw scores but retain hundreds to thousands of textual tokens [3].
On LooGLE-v2 it records the highest raw score of 23.54 and the best CAP of 32.67, again with 16 tokens [3].

Limitations
The fixed 16-token visual prefix may be insufficient for queries that require evidence from multiple distant document segments, and the current evaluation is limited to multiple-choice QA; extending to open-ended generation and adaptive prefix budgets is left for future work [4].

Sources

[1] arXiv 2611.13675v1.pdf, arXiv abstract

[2] arXiv 2611.13675v1.pdf, page 3

[3] arXiv 2611.13675v1.pdf, page 8

[4] arXiv 2611.13675v1.pdf, page 10

[P5] PatternBloom: Empowering Agentic RAG with Externalized RL-Distilled Graph Patterns

Yi Ge, Heng Huang · arXiv 2611.10477v1 · https://arxiv.org/abs/2611.10477v1 · relevance 0.35

TL;DR: PatternBloom introduces a two-stage RL framework that externalizes procedural reasoning as graph pattern memory, achieving state-of-the-art multi-hop QA performance with a 7B model.

Problem
Current retrieval-augmented generation (RAG) systems retrieve only content units (passages, KG fragments, hyperedges) and keep the reasoning route inside the language model's hidden state, causing performance to plateau as reasoning depth increases [1].

Method
PatternBloom shifts retrieval from "what to read" to "how to reason" by (i) using an Information-Density Reward (IDR) to score evidence-graph construction, (ii) distilling high-IDR trajectories (RIDR >= 0.85) into a frozen Graph Pattern Memory (GPM) of 27 type-abstracted reasoning skeletons merged at cosine similarity >= 0.85, and (iii) refining the policy with a Pattern-Augmented Reward (PAR) that combines answer-level F1 and pattern-coverage weighted by ?P = 0.2 [1][2][3].

Key results
- A 7B PatternBloom model outperforms 27 baselines by +15.0 average OOD exact-match (Avg-OOD EM) and beats a 72B prompt-engineered RAG panel by +17.1 Avg-OOD EM [1][3].
- It ranks first on EM and F1 across all seven QA datasets, with an average OOD EM gain of 15.0 over the next best RL method (R1-Searcher) and 11.3 EM gain when transferred to a Qwen3-8B backbone [3].
- Component ablation shows F1 drops from 55.9 (full model) to 44.1 without PAR, to 46.6 without GPM, and to 52.9 without IDR [3].
- The frozen GPM contains 27 skeletons that each aggregate several hundred high-IDR trajectories, and the memory size remains <= 105 patterns with a 4 MB per 1,000-pattern footprint [2][6].

Limitations
The procedural-memory mechanism is evaluated only on a Wikipedia-hyperedge substrate; transfer to other domains such as biomedical, legal, or financial texts is not addressed in this work [7].

Sources

[1] arXiv 2611.10477v1.pdf, arXiv abstract

[2] arXiv 2611.10477v1.pdf, page 6

[3] arXiv 2611.10477v1.pdf, page 7

[6] arXiv 2611.10477v1.pdf, page 18

[7] arXiv 2611.10477v1.pdf, page 37